

Topic 2

Algebra and graphs

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

Learning focus

Explain every method before practising. Use the original practice questions to check calculation, reasoning and communication.

Topic map

2.1	Expressions and algebraic manipulation
2.2	Equations and inequalities
2.3	Sequences and nth terms
2.4	Direct and inverse proportion
2.5	Functions and graphs
2.6	Gradients, area under graphs and algebraic methods

Core foundations and Extended applications are integrated throughout. Study each example, then attempt the practice paper without the notes.

1. Expressions and factorisation

Collect like terms before expanding. Multiply every term in a bracket; factorisation reverses expansion. For $x^2 + bx + c$, find two numbers with sum b and product c . Algebraic fractions follow the same rules as numerical fractions, with excluded values where a denominator is zero.

Worked example

$x^2 + 7x + 12 = (x + 3)(x + 4)$. Check by expanding: $x^2 + 4x + 3x + 12$.

Exam check: $(a + b)^2 = a^2 + 2ab + b^2$, not $a^2 + b^2$.

2. Solving and checking equations

Use inverse operations on both sides and show the line where the variable is isolated. In a simultaneous system, eliminate one variable or substitute an expression for it. A quadratic may factorise; check both roots by substitution. For an inequality, reverse the sign only after multiplying or dividing by a negative number.

Worked example

Solve $2x + y = 11$ and $x - y = 1$. Adding gives $3x = 12$, $x = 4$, then $y = 3$.

Exam check: The question may ask for integer solutions, so distinguish an interval from the integers within it.

3. Sequences and change

For an arithmetic sequence the first difference is constant; n th term = first term + $(n - 1) \times$ difference. For a quadratic sequence, constant second difference suggests a term containing n^2 . A geometric sequence has a constant ratio, so its n th term is first term $\times$ ratio $^{n-1}$.

Worked example

For 5, 9, 13, 17, ... difference = 4 and first term = 5; term n is $5 + 4(n - 1) = 4n + 1$.

Exam check: Test the first and second terms of a proposed n th-term rule.

4. Proportion and functions

In $y \propto x^2$, write $y = kx^2$ and find k from one pair of values. For inverse proportion, $y = k/x$ or k/x^2 as stated. A function maps input to output; evaluate a composite function from the inside out. An inverse function reverses a one-to-one mapping.

Worked example

If $y \propto x^2$ and $y = 45$ when $x = 3$, $k = 45/9 = 5$, hence $y = 5x^2$.

Exam check: Do not use the inverse proportion formula for a direct proportion question.

5. Graphs and rate of change

A straight line $y = mx + c$ has gradient m and vertical intercept c . Solve intersections by equating expressions or reading a graph accurately. The gradient of a distance-time graph is speed; on a speed-time graph, gradient is acceleration and area under the graph is distance.

Worked example

Speed grows uniformly from 0 to 12 m/s over 5 s. Distance = triangular area = $0.5 \times 5 \times 12 = 30$ m.

Exam check: Label both axes and their units; a steep graph means a large gradient, not always a large value.

6. Interpreting curves

A quadratic $y = ax^2 + bx + c$ has a turning point and may cross the x -axis at two roots. For a curve, draw a tangent at the point to estimate the local gradient. Locate solutions to $f(x) = 0$ at x -axis intersections and to $f(x) = g(x)$ where two curves meet.

Worked example

For $y = x^2 - 5x + 6$, factorise as $(x - 2)(x - 3)$, so the x -intercepts are 2 and 3.

Exam check: A numerical graph reading is an estimate; an algebraic solution can give exact roots.

Before the practice paper

Method and accuracy

Write the formula or first operation, show the intermediate step, give units when needed and check by substitution, estimation or a sketch. On non-calculator work, keep exact forms when requested.

Practice paper

Topic 2 · Algebra and graphs

Instructions

Original Cambridge-style practice. Answer all questions. Section A is multiple choice; Section B requires working and final answers. Try Section A without a calculator; use a calculator in Section B only where needed.

Candidate name:	
Class / date:	

Total: 32 marks

Section A · Multiple choice

Choose one answer for each question.

1. Simplify $3x + 2x - 4$. [1]

- A $5x - 4$
- B $6x - 4$
- C $x - 4$
- D $5x + 4$

2. Solve $4x - 7 = 9$. [1]

- A 2
- B 4
- C 8
- D 16

3. Factorise $x^2 - 9$. [1]

- A $(x - 9)(x + 1)$
- B $(x - 3)^2$
- C $(x - 3)(x + 3)$
- D $x(x - 9)$

4. What is the n th term of 2, 5, 8, ...? [1]

- A $2n + 1$
- B $3n - 1$
- C $3n + 1$
- D $n + 2$

5. Find the gradient of $y = -2x + 7$. [1]

- A -2
- B 2
- C 7
- D -7

6. For $y = 3x^2$, find y when $x = -2$. [1]

- A -12
- B -6
- C 6
- D 12

7. A speed-time graph has a constant positive gradient. What is constant? [1]

- A speed
- B distance
- C acceleration
- D time

8. Solve $x^2 - 5x + 6 = 0$. [1]

- A -2,-3
- B 2,3
- C 1,6
- D -1,-6

Section B · Structured questions

9. A rectangle has length $x + 5$ cm and width $x - 1$ cm.

(a) Expand an expression for its area. [2]

(b) If $x = 6$, calculate the area. [2]

(c) Factorise $x^2 + 4x - 5$ to check your expression. [2]

[Total: 6]

10. The line $3x + 2y = 16$ intersects the line $x - y = 2$.

(a) Rearrange $x - y = 2$ to make y the subject. [1]

(b) Solve the pair of simultaneous equations. [3]

(c) Check the point in $3x + 2y = 16$.

[1]

[Total: 5]

11. A taxi costs \$4 plus \$1.80 per kilometre.

(a) Write a formula C in terms of distance d .

[2]

(b) Find the cost for 12 km.

[2]

(c) Find d when the cost is \$31.

[2]

[Total: 6]

12. A ball has height $h = -5t^2 + 20t$ (metres) after t seconds.

(a) Find h when $t = 1$.

[2]

(b) Find the two times when $h = 0$.

[2]

(c) Find its maximum height by completing the square or symmetry.

[3]

[Total: 7]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Award method marks when the mathematical setup is correct even if an arithmetic slip follows.
Check each part independently.

Section A · Answers and reasons

1. A — $5x - 4$. Only like x-terms combine: $3x + 2x = 5x$.
2. B — 4. Add 7, then divide by 4: $x = 4$.
3. C — $(x - 3)(x + 3)$. Difference of squares: $x^2 - 3^2 = (x-3)(x+3)$.
4. B — $3n - 1$. The difference is 3; $3n - 1$ gives first term 2.
5. A — -2. The x coefficient in $y = mx + c$ is the gradient.
6. D — 12. Squaring -2 gives 4, then multiply by 3.
7. C — acceleration. The slope on a speed-time graph is the change of speed per unit time.
8. B — 2,3. $(x-2)(x-3) = 0$ gives two roots.

Section B · Worked solutions

9. a) $(x + 5)(x - 1) = x^2 + 4x - 5 \text{ cm}^2$ [2]. b) $(11)(5) = 55 \text{ cm}^2$ [2]. c) $(x + 5)(x - 1)$ [2].

10. a) $y = x - 2$ [1]. b) $3x + 2(x - 2) = 16$, so $5x = 20$, $x = 4$, $y = 2$ [3]. c) $3(4) + 2(2) = 16$ [1].

11. a) $C = 4 + 1.8d$ [2]. b) $C = 4 + 1.8(12) = \$25.60$ [2]. c) $31 = 4 + 1.8d$, $d = 27/1.8 = 15 \text{ km}$ [2].

12. a) $-5 + 20 = 15 \text{ m}$ [2]. b) $-5t(t - 4) = 0$, so 0 and 4 s [2]. c) Vertex is halfway at $t = 2$; $h(2) = -20 + 40 = 20 \text{ m}$ [3].

Review your errors

Mark the first wrong step: setup, algebra, arithmetic, units, graph or reasoning. Redo that part without seeing the solution.

Syllabus reference: Cambridge IGCSE Mathematics 0580, 2025-2027,
www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf. Paper style informed by the June 2026 papers and
examiner report provided by the user. All practice questions are original.